RIPHAH INTERNATIONAL UNIVERSITY

DATA STRUCTURES LAB

Lab Task 12

Name: Aroosa Feroz

SAP ID: 68997

Submitted to: Ma'am Nehan Sahar

Subject: Data Structures Lab

DEPARTMENT OF COMPUTER SCIENCE

RIPHAH INTERNATIONAL UNIVERSITY, GULDBERG GREENS CAMPUS ISLAMABAD

Task 1: Identify Tree Properties

Given a tree:

```
        A
      /   \
    B       C
   / \       \
  D   E       F
```

Find:

- Root node
- Leaf nodes
- Degree of B
- Height of tree
- Level of node E

Ans:

Root Node: Root is the topmost node having no parent. Here in tree Root node is A.

Leaf Nodes: Leaf nodes are those with no children. Here D,E,F

Degree of B: Degree = number of children of a node for B it is 2 (Children: D and E)

Height of tree: Longest path from root to any leaf here it is 2

Paths:

- A → B → D = 2 edges
- A → B → E = 2 edges
- A → C → F = 2 edges

Maximum = 2 so height is 2.

Level of node E: Level starts from 0 at root

Levels:

- Level 0 → A
- Level 1 → B, C
- Level 2 → D, E, F

Here for E it is 2

Thank You